

The flavor off-diagonal coupling constants f_{ij} and g_{ij} induce lepton flavor violation (LFV). From the results of $\mu \rightarrow e\gamma$, $\tau \rightarrow e\gamma$ and $\tau \rightarrow \mu\gamma$, $|f_{\mu\tau}f_{\tau e}|$, $|f_{\tau\mu}f_{\mu e}|$ and $|f_{\tau e}f_{e\mu}|$ are respectively constrained as a function of m_ω . The data of rare decays of $\mu \rightarrow eee$, $\tau \rightarrow \mu\mu e$ and $\tau \rightarrow \mu ee$ are also used to constrain the combinations $|g_{\mu e}g_{ee}|$, $|g_{\tau e}g_{\mu\mu} + g_{\tau\mu}g_{\mu e}|$ and $|g_{\tau e}g_{\mu e} + g_{\tau\mu}g_{ee}|$, respectively, depending on m_k . The $g-2$ data can also be used to constrain a combination of these coupling constants with m_ω and m_k ¹.

In the scenario with hierarchical neutrino masses, f_{ij} satisfy $f_{e\mu} \simeq f_{e\tau} \simeq f_{\mu\tau}/2$. The typical relative magnitudes among the coupling constants g_{ij} can be $g_{\mu\mu} : g_{\mu\tau} : g_{\tau\tau} \simeq 1 : m_\mu/m_\tau : (m_\mu/m_\tau)^2$. For $g_{\mu\mu} \simeq 1$, the neutrino data and the LFV data give the constraints such as $m_k \gtrsim 770$ GeV and $m_\omega \gtrsim 160$ GeV [10]. On the other hand, the constraints on the couplings and masses are more stringent for the inverted neutrino mass hierarchy. The current data then gives $m_\omega \simeq 825$ GeV for $g_{\mu\mu} \simeq 1$ [10]. One of the notable things in this case is the lower bound on $\sin^2 2\theta_{13}$, which is predicted as around 0.002 [9].

B. Models with TeV-scale Right-handed Neutrinos with a discrete Z_2 symmetry

Similar to the tree-level seesaw model, tiny masses of left-handed neutrinos would also come from Majorana masses $M_{N_R^\alpha}$ of gauge-singlet right-handed neutrinos N_R^α in the radiative seesaw scenario [4–6]. One simple way to realize the absence of the tree-level Yukawa interaction $\bar{\nu}_L^i \tilde{\Phi} N_R^\alpha$ is introduction of a discrete Z_2 symmetry, with the assignment of the odd quantum number to N_R^α and the even to the SM particles. To obtain the dimension five operator in Eq. (1) at the loop level, we need to introduce additional Z_2 -odd scalar fields. The lightest of all the Z_2 -odd particles can be a candidate of DM if it is electrically neutral. The original model of the radiative seesaw model with such a discrete symmetry is proposed by Krauss, Nasri, and Trodden [4], in which neutrino masses are induced at the three-loop level. In the following, we consider two variant models of the Kraus-Nasri-Trodden (KNT) model.

¹ If we take $g_{ee} = 0$, then m_k is unbounded from the $\mu \rightarrow eee$ and $\tau \rightarrow \ell ee$ results ($\ell = e$ or μ), so that relatively light $k^{\pm\pm}$ ($m_k \sim 100 - 200$ GeV) are possible.